

MADHAVA MATHEMATICS COMPETITION 2024

(A Mathematics Competition for Undergraduate Students)

Organized by: Bhaskaracharya Pratishthana, Pune

Time: 12.00 noon to 3.00 p.m.

Maximum Marks: 100

*N.B.: Part I carries 20 marks, Part II carries 30 marks, and Part III carries 50 marks.***PART I (10 Questions $\times$ 2 Marks = 20 Marks)**

- Let $a < b < c$ be three positive integers in geometric progression whose common ratio is an integer. If the arithmetic mean of a, b, c is $b + 2$, then the value of a is:
(A) 24 (B) 12 (C) 6 (D) 4
- Triangles are formed by joining the vertices of a regular polygon of 12 sides. The number of triangles having no side in common with the polygon is:
(A) 124 (B) 220 (C) 96 (D) 112
- If $1, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are the n -th ($n > 2$) roots of unity, then $\sum_{k=1}^{n-1} k\alpha_k$ equals:
(A) $\frac{n}{2}$ (B) $\frac{n(n-1)}{2}$ (C) $-\frac{n}{1-\alpha}$ (D) $-\frac{n}{2}$
- If $f(x) = \begin{cases} ax + b, & x < -1 \\ 4x^2 + 1, & -1 \leq x \leq 1 \\ 2x - 1, & x > 1 \end{cases}$ is continuous everywhere, then which value can ab never take?
(A) -3 (B) -1 (C) 0 (D) 3
- Let $f : [1, 3] \rightarrow \mathbb{R}$ be a continuous function. If $f(x)$ takes rational values for all x and $f(2) = 10$, then $f(1.5) =$:
(A) 7.5 (B) 10 (C) 1.5 (D) 5
- If $f(x) = (x + 1)^2 - 1$ for $x \geq -1$, then the set $\{x : f(x) = f^{-1}(x)\}$ is:
(A) $\{0\}$ (B) $\{0, -1\}$ (C) $\mathbb{R}$ (D) $\emptyset$
- The number of real solutions of $1 + |x - 1| = e^x(x - 2)$ is:
(A) 0 (B) 1 (C) 2 (D) 3
- In how many ways can you choose an odd number of objects from n objects?
(A) 2^{n-1} (B) 2^n (C) 2^{n+1} (D) n
- The family of circles with center $(a, 0)$ and radius a satisfies the differential equation:
(A) $2xyy' + y^2 - x^2 = 0$ (B) $2xyy' + x^2 - y^2 = 0$ (C) $2xy + x - y = 0$ (D) $2xy' + x^2 + y = 0$
- The series $\sum_{n=1}^{\infty} \frac{nx \cos(nx) + \sin(nx) - e^{-|x|}}{n^3 + 1}$:
(A) converges pointwise everywhere on $\mathbb{R}$, but not uniformly (B) does not converge at any point (C) converges uniformly on $\mathbb{R}$ (D) converges only at 0

PART II (5 Questions $\times$ 6 Marks = 30 Marks)

1. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function with $\int_0^1 x(x-1)^2 f(x) dx = 0$. Prove that there exists $c \in (0, 1)$ such that $\int_0^c x^2 f(x) dx = c \int_0^c x f(x) dx$.
2. Let $f(x) = x^2 + 4x + 3$ and $g(x) = 2x + 5$. Show that $\gcd(f(n), g(n)) = 1$ or 3 as n varies over integers.
3. Let $a_1, a_2, \dots, a_n$ be real numbers. Prove that $\sum_{i,j=1}^n e^{a_i a_j} \geq 0$.
4. Suppose for $n \geq 4$, $x_1, x_2, \dots, x_n$ are non-negative integers such that after removing any one of them, the remaining numbers can be partitioned into two sets with equal sums. (i) If n is even, prove $x_i = 0$ for all i . (ii) Does the same hold when n is odd?
5. Give an example of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ which is continuous at a if and only if a is a positive integer.

PART III (4 Problems = 50 Marks)

1. Suppose α, β are roots of $x^2 + bx + c$. Let $p \geq 2$ be a prime such that $p \mid b, p \mid c$, but $p^2 \nmid c$. (i) Find largest power of p dividing $\alpha^2 + \beta^2$. [2] (ii) Find largest power of p dividing $\alpha^3 + \beta^3$. [3] (iii) For any n , find largest power of p dividing $\alpha^{2^n} + \beta^{2^n}$. [8]
2. (i) Show that $x^6 - 6x^5 + 1 = 0$ has a unique root in $(0, 1)$. [3] (ii) Show that every complex non-real root z satisfies $|z| < z_0$, where z_0 is the root in $(0, 1)$. [10]
3. (i) Suppose A, B are $n \times n$ complex matrices with $AB^2 - B^2A = B$. Show B is nilpotent. [9] (ii) Does there exist A for a given nilpotent B such that $AB^2 - B^2A = B$? [3]
4. Let A, B, C, D be vertices of a quadrilateral inscribed in the unit circle, and P be inside $ABCD$. Show that $|PA||PB||PC||PD| \leq \frac{27}{16}$. [12]